



Inspiring Excellence

Department of Electrical and Electronic Engineering
Hardware / Software Report Submission

Semester: Summer 2026

Course Code: EEE101L

Course Title: Electrical Circuits I Laboratory

Section: 03

Experiment No: 03

Experiment Name: Verification of Superposition Principle

Group No: 01

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Date of Submission: 06/07/2026

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Electrical Circuits I Laboratory
EEE101L

Department of Electrical & Electronic Engineering (EEE)
BRAC University

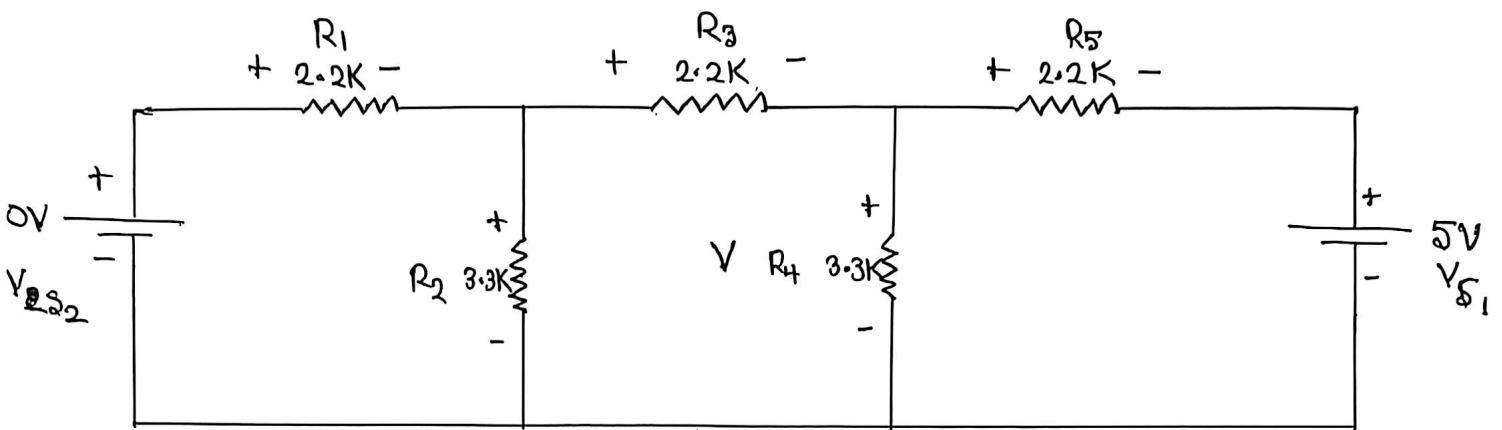
Experiment No. 3

Verification of Superposition Principle

1. Objective: To verify experimentally the superposition theorem which is an analytical technique of determining currents in a circuit more than one emf source.

2. Theoretical Background: In a linear circuit containing multiple independent sources and linear elements the voltage across (or the current through) any element when all the sources are acting simultaneously may be obtained by adding algebraically all the individual voltages (or the currents) caused by each independent source acting alone, with all other sources deactivated.

An independent voltage source is deactivated (made zero) by shorting it and an independent current source is deactivated (made zero) by open circuiting it. However, if a dependent source is present, it must remain active during the superposition process.



According to superposition principle:

$$V_{\text{total}} = V_1 + V_2 \quad \left\{ \begin{array}{l} V_1 = 5V \text{ short} \\ V_2 = 10V \text{ short} \end{array} \right.$$

3. Equipment:

- ⇒ Two DC power supplies adjusted to 10V and 5V
- ⇒ Resistors: 2.2K, 3.3K, 2.2K, 3.3K and 2.2K
- ⇒ Bread board / Trainer board
- ⇒ One digital Multimeter (DMM)

4. Circuit Diagram:

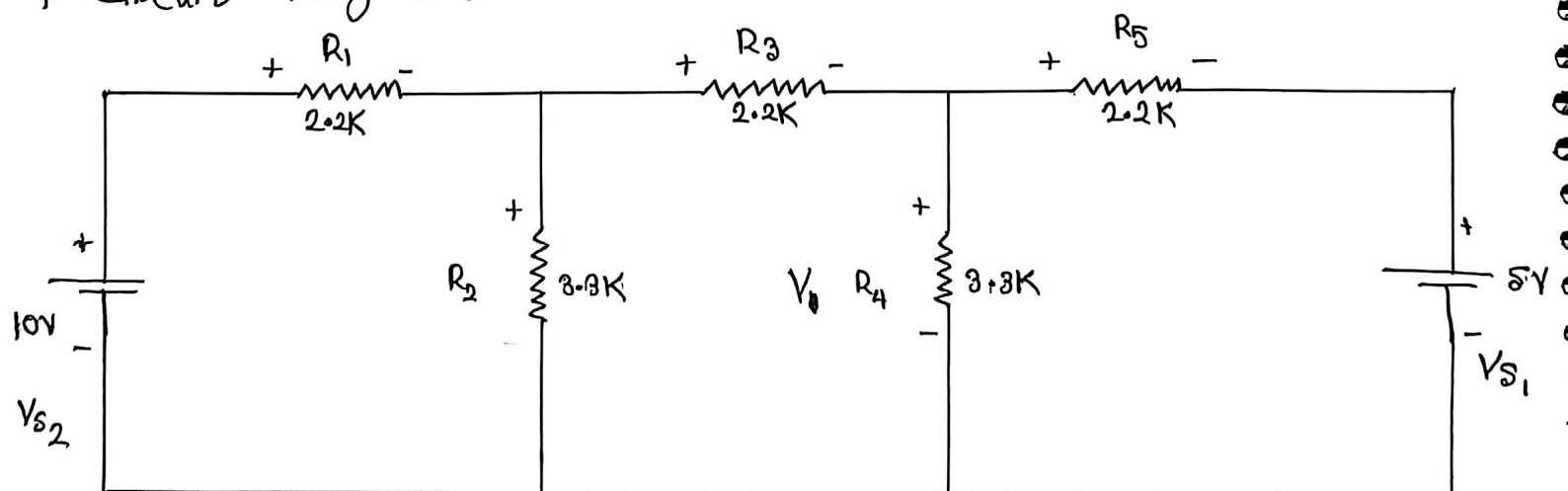


Fig: 01

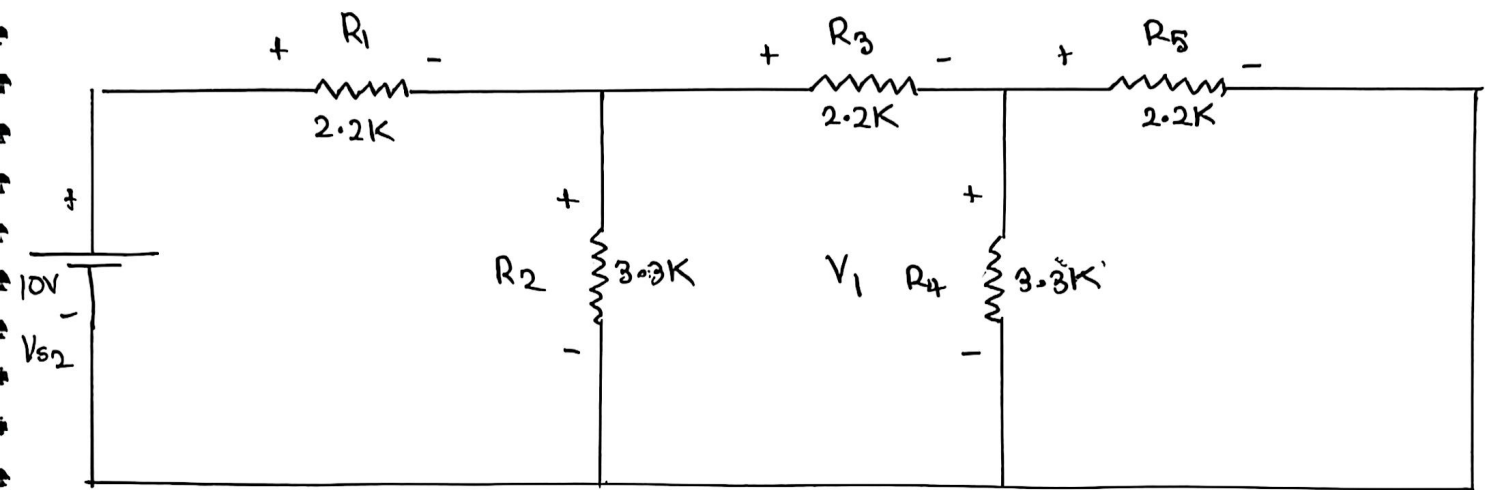


Fig: 02 [V_{s1} off]

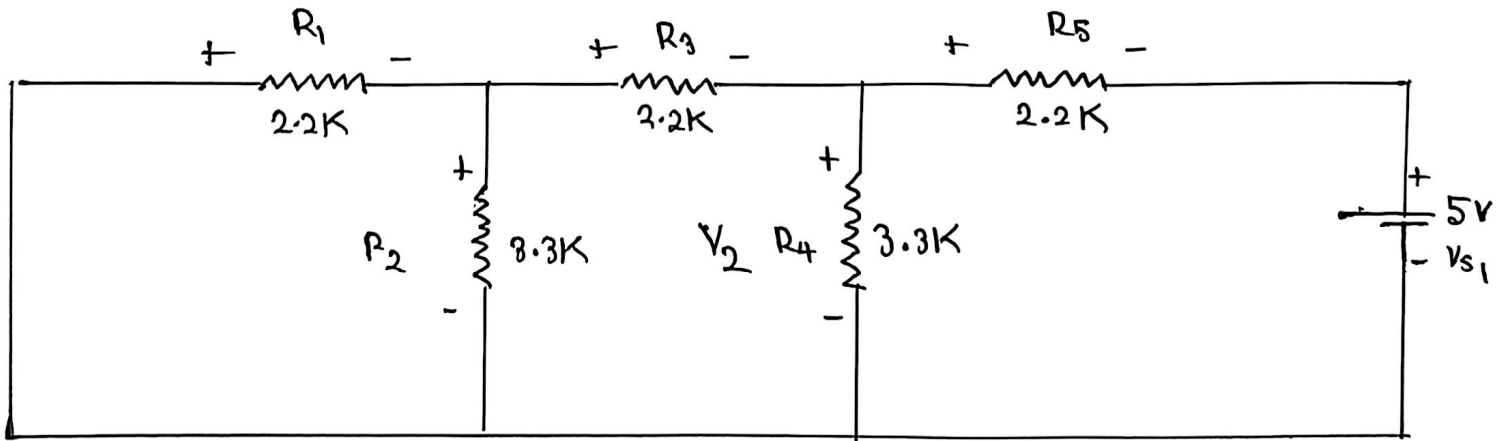


Fig: 03 [V_{s2} off]

5. Data Table :

Fig 01:-

R_4 (K- Ω)	V (v)
3.26	3.86

Fig 02 :

R_4 (K Ω)	V_1 (V)
3.86	1.63

Fig 03 :

R_4 (K Ω)	V_2 (V)
3.86	2.23

Verification :

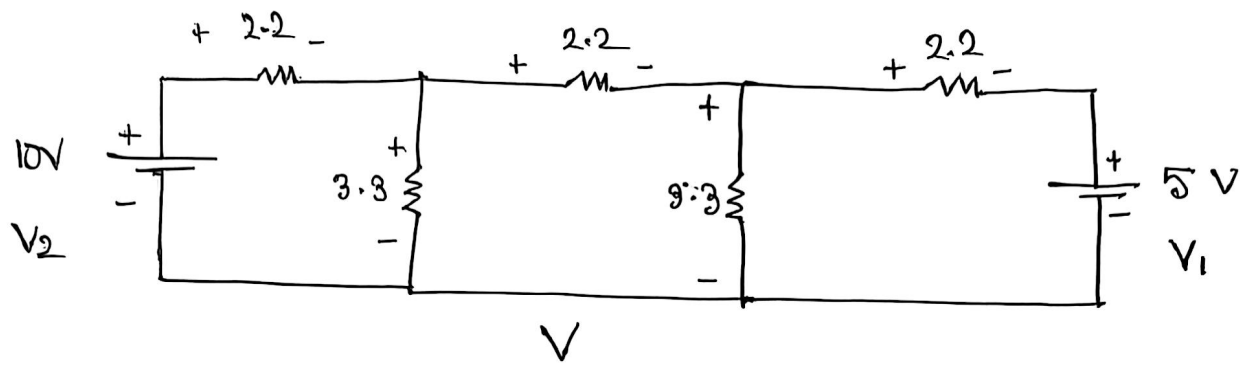
V (V)	$V_1 + V_2$
3.86	3.86

6. Question Answer:

The superposition Theorem is fundamentally rooted in the mathematical property of linearity. $V=IR$ and $I=V/R$ are linear equations, they can be measured by the superposition principle. But on the other hand, $P=I^2R$ or $P=V^2/R$ is a quadratic function, we cannot directly measure them by the principle, rather we will get wrong value. So superposition principle cannot be used to determine power.

1.42

Exp-3



અભાવ $\Rightarrow V = V_1 + V_2$

$V = 3.86V$

$V_1 = 1.63V$
 $V_2 = 2.23$ } $= 3.86V$

$\left(\begin{matrix} V_1 = 5V \text{ off} \\ V_2 = 10V \text{ off} \end{matrix} \right)$

A handwritten signature in blue ink, consisting of a large, sweeping 'V' shape with a loop at the bottom.